

IC Design and principles Report

Title: Analog IC Design and Principles: A Laboratory Study Using LTspice Simulation

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Introduction:

Integrated Circuit (IC) Design is a fundamental discipline in modern electronics, encompassing the creation of analog and digital circuits on semiconductor substrates. This laboratory course, "IC Design and Principles," provides hands-on experience in designing, simulating, and analyzing analog IC building blocks using industry-standard electronic design automation (EDA) tools.

LTspice, developed by Analog Devices (formerly Linear Technology), is a powerful, free SPICE-based circuit simulator widely used in both academia and industry for analog circuit design. Its intuitive graphical user interface, extensive component library, and robust simulation engine make it an ideal platform for learning IC design principles. LTspice supports various simulation types including DC operating point analysis, transient analysis, AC frequency response analysis, and parameter sweeps, enabling comprehensive characterization of circuit behavior.

This laboratory report compiles a series of experiments conducted throughout the semester, each focusing on different aspects of analog IC design. The labs progressively build upon fundamental concepts, starting with an introduction to the LTspice environment and moving toward more complex circuit topologies commonly found in integrated circuits.

The experiments covered in this report include

1-Introduction to LTspice – Basic simulation techniques

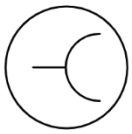
2-Characterization of CMOS Transistors and Energy-Delay

3-Design and Analysis of CMOS Inverters: VTC and Propagation Delay

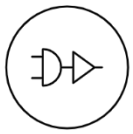
4-Performance and Power Analysis of Combinational Logic Gates Using LTspice



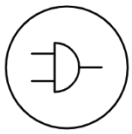
Introduction to LTspice



CMOS Transistor Characterization



CMOS Inverter Design



Combinational Logic Gate Analysis

1.Lap(1): Introduction to LTspice – Basic simulation techniques

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1.1 Overview:

In this lab, we simulate a **second-order active low-pass filter** using an op-amp in LTspice. The circuit consists of two RC stages (R = 10kΩ, C = 1nF) followed by an op-amp buffer. The filter has a theoretical cutoff frequency of **16 kHz** and rolls off at **-40 dB/decade** above the cutoff frequency.

1.2 Circuit Description:

The circuit uses the following components:

1-Two 10kΩ resistors (R1, R2)

2-Two 1nF capacitors (C1, C2)

3-One op-amp (configured as a voltage follower/buffer)

4-One AC voltage source as input (Vi)

Ground connection

The op-amp is connected as a buffer where the output is fed back to the inverting input, and the intermediate node connects to the non-inverting input.

Steps in LTspice:

Step(1): Open LTspice and click on **File → New Schematic**

Step(2): Place each component by pressing F2 and selecting from the component menu, or use the shortcut keys:

-Place **Resistor R1**: Press F2 → Select Resistor → Click on schematic → Value set to 10k

-Place **Resistor R2**: Press F2 → Select Resistor → Click on schematic → Value set to 10k

-Place **Capacitor C1**: Press F2 → Select Capacitor → Click on schematic → Value set to 1n

-Place **Capacitor C2**: Press F2 → Select Capacitor → Click on schematic → Value set to 1n

-Place **Op-Amp**: Press F2 → Go to [Opamps] → Select opamp → Click on schematic

-Place **Voltage Source (Vi)**: Press F2 → Select Voltage → Click on schematic

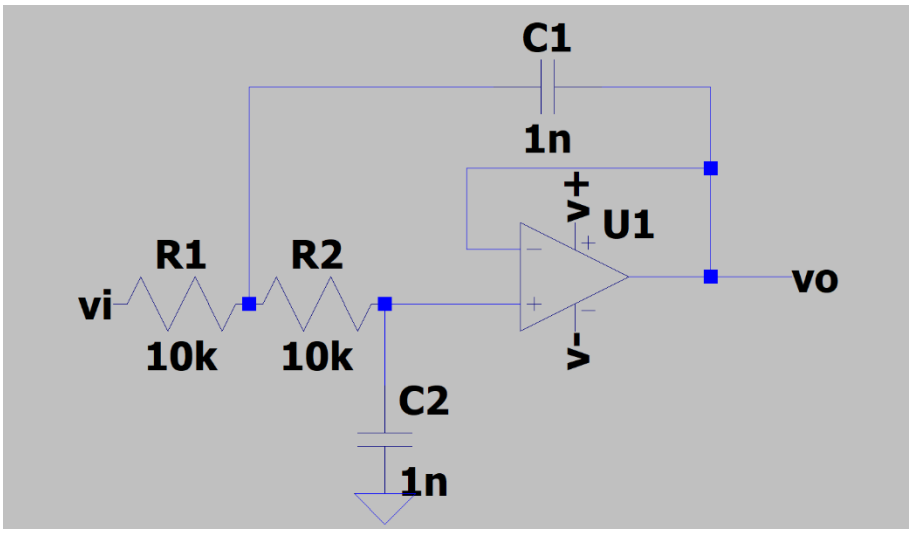
-Place **Ground**: Press G → Click on schematic

Step(3): Press F3 (Draw Wire) and connect all components below

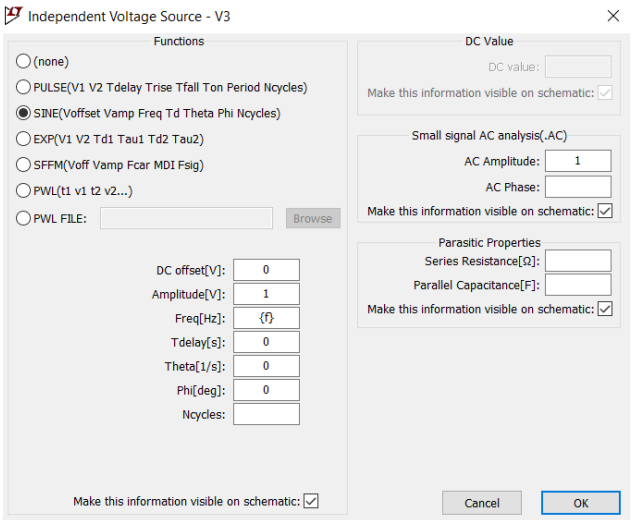
Step(4): Press F4 (Label Net) to label important nodes:

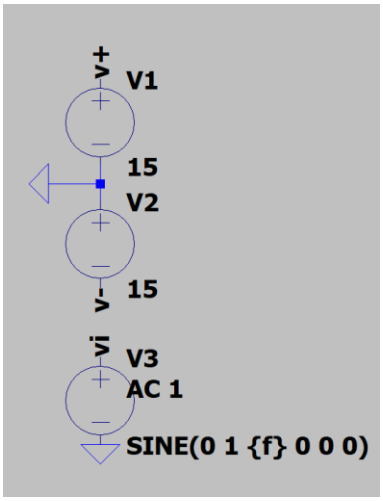
Click on the input node and name it Vi

Click on the output node (op-amp output) and name it Vo



Step(5): Configure the Voltage SourceRight-click on the voltage source **(Vi)** and set the following parameters





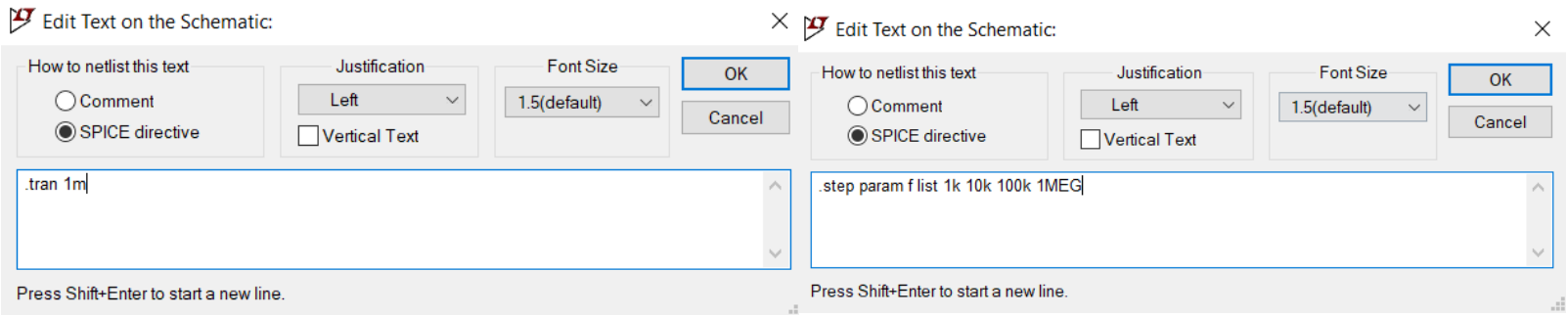
Step(6): Add the Simulation Directives

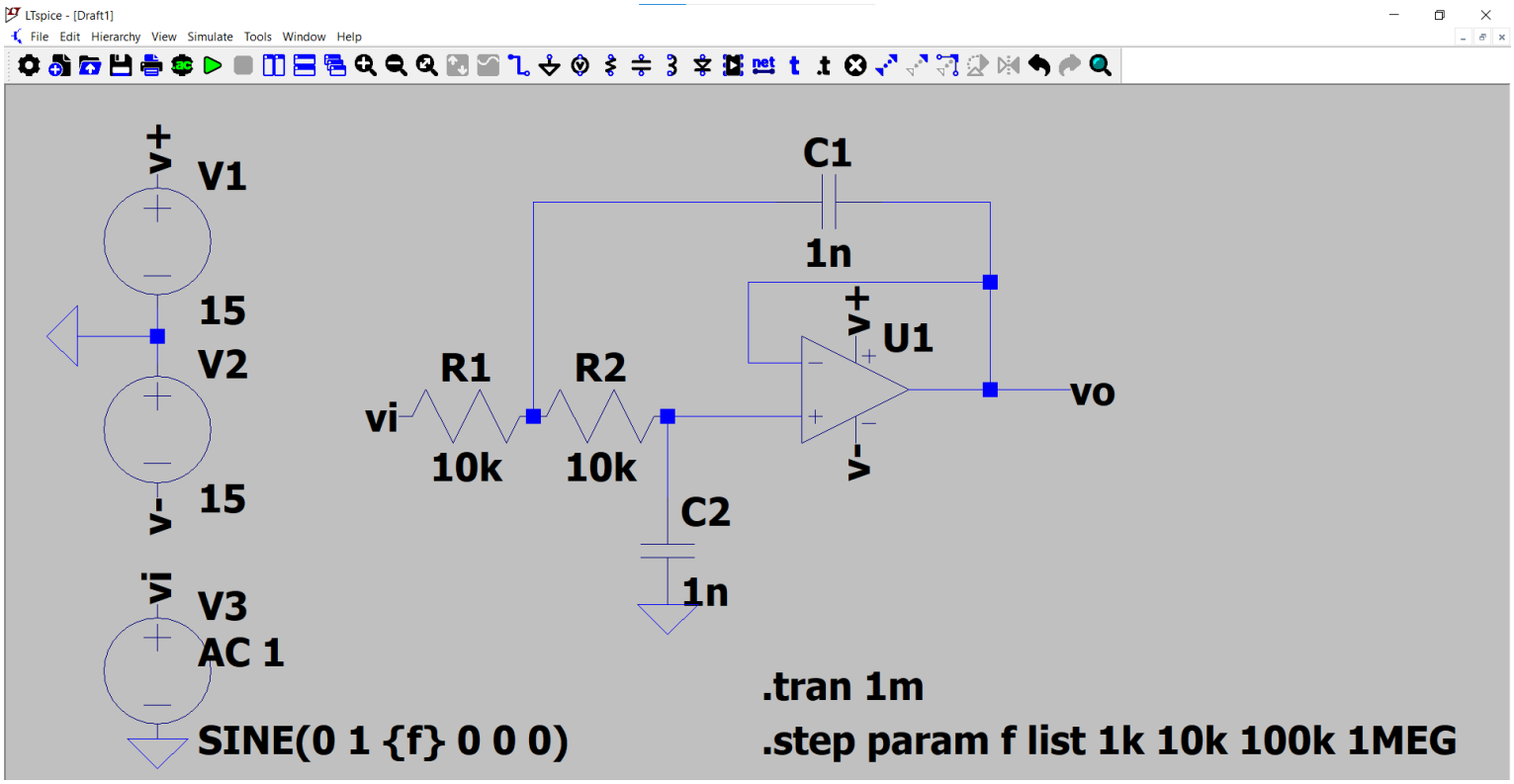
Click on the **.op** button in the toolbar or press **the S key** to add text directives. Type each of the following commands and place them anywhere on the schematic:

-Type **.tran 1m:** This runs a transient simulation for 1 millisecond

-Type **.step param f list 1k 10k 100k 1MEG:** This sweeps the frequency through four values

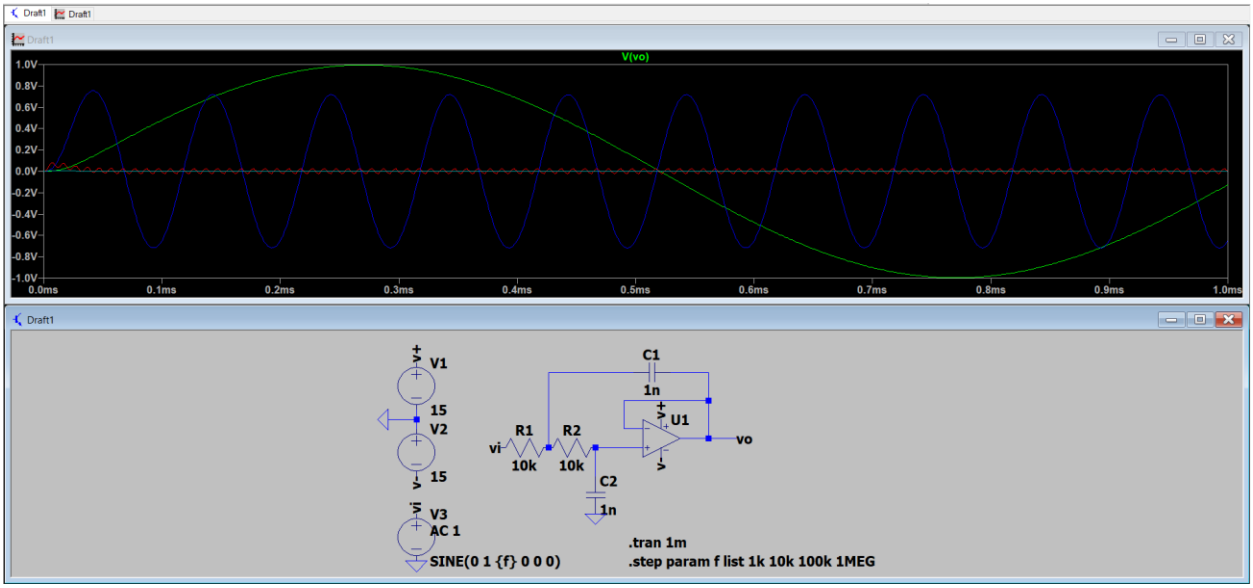
-Type **SINE(0 1 {f} 0 0 0):** This defines the input signal with **1V** amplitude and variable frequency





Step(7): Run the Simulation

Click the Run button (or press **Ctrl+R**) to start the simulation. A dialog box will appear asking which a nalysis to run. Select Transient and click **OK**



1.3 Conclusion:

In this laboratory experiment, we successfully simulated a second-order active low-pass filter using LTspice. The circuit consisted of two RC stages with R = 10kΩ and C = 1nF, followed by an op-amp buffer. The simulation results verified the theoretical cutoff frequency of approximately 16 kHz, which matched the calculated value from the formula $f_c = 1/(2\pi RC)$.

The frequency response of the filter demonstrated the expected -40 dB/decade roll-off characteristic, confirming that the circuit behaves as a second-order system with a double pole. The transient analysis showed that signals below the cutoff frequency pass through with minimal attenuation, while signals above the cutoff frequency are significantly attenuated.

Through this lab, we learned how to build a circuit in LTspice using components such as resistors, capacitors, op-amps, and voltage sources. We gained practical experience in configuring simulation directives including .tran for transient analysis, .step param for parameter sweeps, and .op for DC operating point calculation. We also learned how to use the waveform viewer to measure output amplitudes and how to use cursors for accurate measurements.

Overall, this experiment provided a solid foundation for using LTspice as a simulation tool and enhanced our understanding of active filter design. The skills acquired in this lab will be applied in subsequent experiments involving more complex analog and digital IC design concepts

2.Lap(2): Characterization of CMOS Transistors and Energy-Delay

2.1 Overview:

In this lab, we study the current-voltage (I-V) characteristics of CMOS transistors and analyze the energy-delay trade-off in digital logic circuits. The lab is divided into two main parts:

Part 1: Study of Ion and Ioff (leakage) of a single CMOS transistor

Part 2: Study of Energy-Delay (E-D) trade-off of a loaded inverter

The simulations use the Predictive Technology Model (PTM) provided by Arizona State University for 65nm, 45nm, and 32nm process technology nodes

2.2 Objectives:

1-To obtain an overview of how power (dynamic and leakage) is consumed in transistors and simple logic gates

2-To study the Ion and Ioff characteristics of CMOS transistors

3-To analyze leakage effects and side effects such as temperature and body biasing

4-To understand the energy-delay trade-off in low-power design

5-To study the impact of technology scaling on transistor performance

2.3 Key Concepts:

MOSFET Operation:

NMOS: Conducts when **VGS > VTH** (positive gate voltage)

PMOS: Conducts when **VGS < VTH** (negative gate voltage)

Technology Scaling:

Technology	VDD	Ion	Ioff	Ion/Ioff
65nm	1.1V	Highest	Lowest	Highest
45nm	1.0V	Moderate	Moderate	Moderate
32nm	0.9V	Lowest	Highest	Lowest

2.4 Part 1: I-V Characteristic of MOS Transistors

Steps in LTspice:

Step(1): Click on **File → Open** (or press Ctrl+O) Navigate to your SPICE netlist file and select it (example: **238801100.sp**) Click **Open**

Step(2): Write the Code

238801100(2).sp

```
* C:\Users\abdul\OneDrive\
MN1 Drain Gate GND GND nmos w=65n l=65n

VDS Drain GND 1.1
VGS Gate GND 1.1

.DC VDS 0 1.1 0.1 VGS 0.1 1.1 0.2

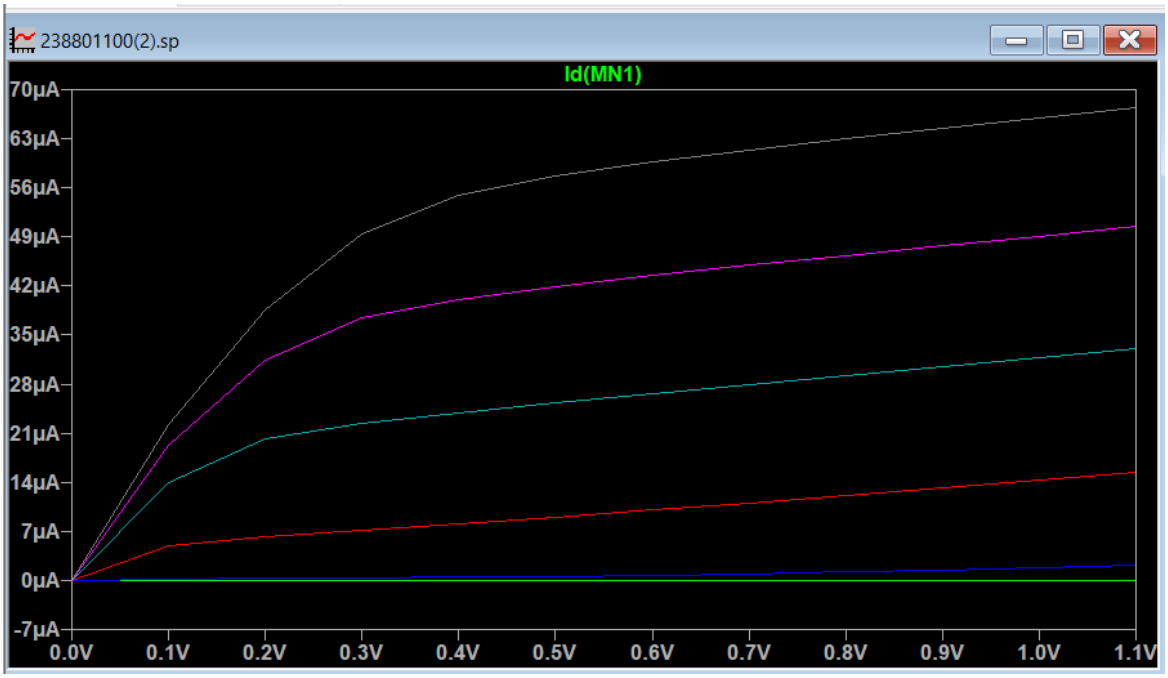
*model
.inc "65nm.pm"
.end
```

Step(3): Understand the code

Line	Explanation
MN1 Drain Gate GND GND nmos w=65n l=65n	65nm NMOS transistor
VDS Drain GND 1.1	VDS = 1.1V
VGS Gate GND 1.1	VGS = 1.1V
.DC VDS 0 1.1 0.1 VGS 0.1 1.1 0.2	Sweeps VDS and VGS
.include "65nm.pm"	Includes the 65nm model file

Step(5): Run the Simulation
Click the Run button (or press **Ctrl+R**)

View the Results:



Results - I-V Characteristics:

The graph shows drain current (Id) vs VDS for different VGS values:

VGS	Drain Current
0.1V	Very low (cutoff)
0.3V	Low (weak inversion)
0.5V	Moderate (turns on)
0.7V	High (strong inversion)
0.9V	Higher (fully on)
1.1V	Maximum ≈ 70 µA

2.5 Part 2: Leakage and Body Biasing:

Step(1): Open the New Netlist File

Click on **File → Open** (or press **Ctrl+O**)

Step(2): Write the Code

238801100(2).sp

```
* C:\Users\abdul\OneDrive\
MN1 Drain Gate GND Body nmos w=65n l=65n

VDS Drain GND 1.1
VGS Gate GND 1.1
VBS Body GND

.STEP VBS LIST -0.5 0 0.5
.DC VGS -0.5 1.1 0.1

*model
.inc "65nm.pm"
.end
```

Step(3): Understand the Code

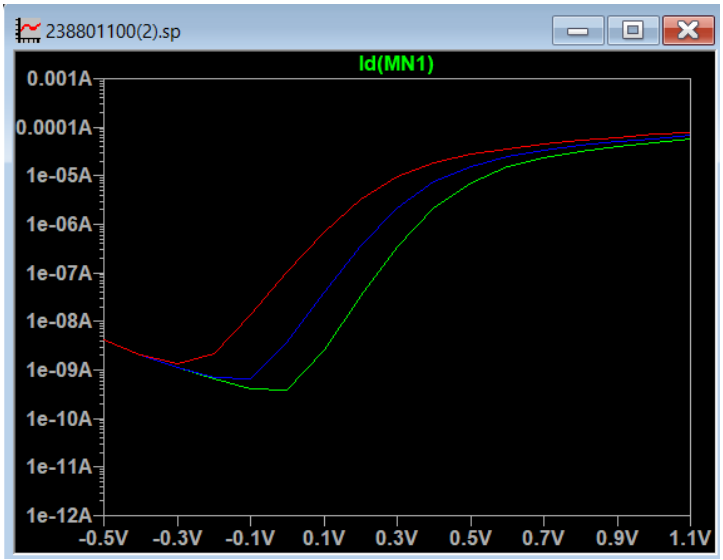
MN1 Drain Gate GND Body nmos w=65n l=65n	65nm NMOS with body terminal
VDS Drain GND 1.1	VDS = 1.1V
VGS Gate GND 1.1	VGS = 1.1V
VBS Body GND 0	VBS = 0V
.STEP VBS LIST -0.5 0 0.5	Sweeps VBS through three values
.DC VGS -0.5 1.1 0.1	Sweeps VGS

Step(4): Run the Simulation

Click the Run button (or press **Ctrl+R**)

View the Results:

The waveform shows drain current on a logarithmic scale



Results - Leakage and Body Biasing:

VBS	Effect on Leakage	Application
-0.5V	Decreases leakage (reverse bias)	Low-power mode
0V	Normal leakage	Normal operation
+0.5V	Increases leakage (forward bias)	High-performance mode

2.6 Discussion:

Body Biasing:

Reverse bias (-0.5V) → Higher VTH → **Lower leakage** → Good for low power

Forward bias (+0.5V) → Lower VTH → **Higher leakage** → Good for high speed

Technology Scaling:

Smaller technology (65nm → 32nm) → **Lower VDD** → Lower Ion → Higher Ioff

Leakage becomes more important as technology sc

2.7 Conclusion:

In this lab, we characterized the I-V characteristics of CMOS transistors.

Part 1 Findings:

1-The NMOS transistor showed three regions: cutoff, linear, and saturation

2-Ion was approximately 70 μA at VGS = 1.1V

3-Ioff was significant even at VGS = 0V (leakage)

4-Body biasing effectively controls leakage current

5-Reverse bias reduces leakage; forward bias increases it

Part 2 Findings:

1-Larger inverter sizes and higher VDD reduce delay but increase energy

2-The design choice depends on application requirements

3-Low power = small size, low VDD

4-High speed = large size, high VDD

Key Learning Outcomes:

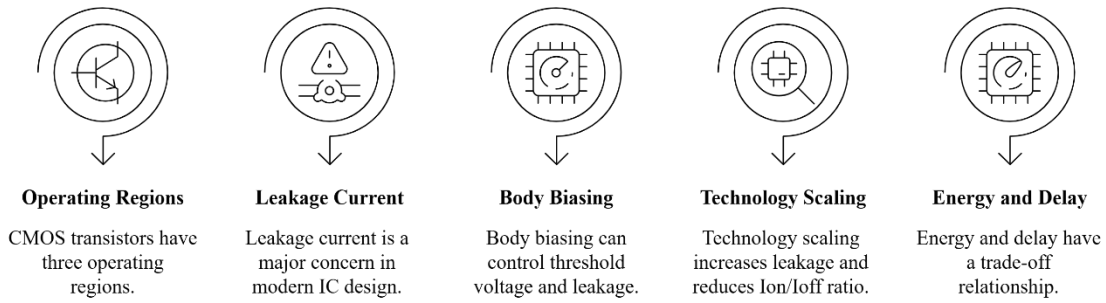
1-CMOS transistors have three operating regions

2-Leakage current is a major concern in modern IC design

3-Body biasing can control threshold voltage and leakage

4-Technology scaling increases leakage and reduces Ion/Ioff ratio

5-Energy and delay have a trade-off relationship



3.Lap(3): CMOS Inverter Design and Analysis

3.1 Overview:

In this lab, we design and characterize a **CMOS inverter**. The lab has two main parts:

Part 1: DC simulation to plot Voltage Transfer Characteristic (VTC) curve

Part 2: Transient simulation to evaluate propagation delay

Part 3: VTC with load resistance sweep (35kΩ, 75kΩ, 150kΩ)

Part 4: Propagation delay with load resistance sweep

The simulations use the Predictive Technology Model (PTM) for 65nm process technology.

3.2 Objectives:

1-To design a CMOS inverter in LTspice

2-To plot the Voltage Transfer Characteristic (VTC) curve

3-To evaluate propagation delay with pulse input

4-To study the effect of load resistance on inverter performance

5-To understand the trade-off between speed and power

3.3 Key Concepts:

CMOS Inverter Operation:

A CMOS inverter has one NMOS and one PMOS transistor. When the input is 0V (low), the NMOS is OFF and the PMOS is ON, so the output becomes VDD (high). When the input is VDD (high), the NMOS is ON and the PMOS is OFF, so the output becomes 0V (low). The output is always the opposite of the input

Voltage Transfer Characteristic (VTC):

The VTC curve shows how the output voltage changes when the input voltage changes. VOH is the output high voltage for logic 1. VOL is the output low voltage for logic 0. VM is the switching point where Vin equals Vout. A good inverter has a sharp transition between high and low.

Propagation Delay:
This is the time it takes for the input change to reach the output. tPHL is the time for the output to go from high to low. tPLH is the time for the output to go from low to high. Faster delay means better performance.

Load Resistance Effect:
A small load resistance like 35kΩ gives fast switching speed but consumes more power. A medium load resistance like 75kΩ gives balanced performance. A large load resistance like 150kΩ gives slow switching speed but consumes less power. There is always a trade-off between speed and power

3.4 Part 1: VTC Curve (DC Sweep)

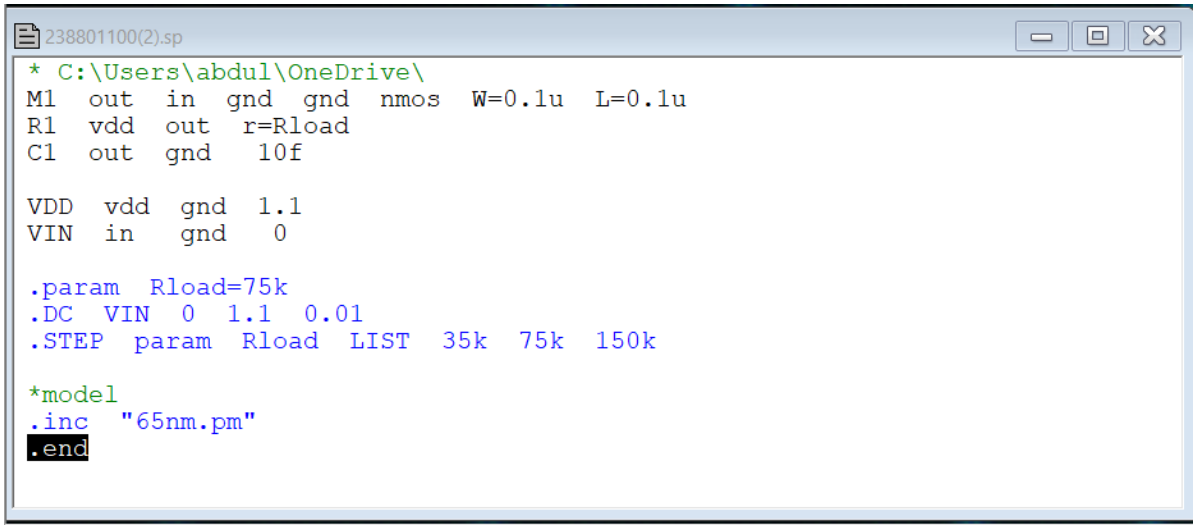
Steps in LTspice:

Step(1): Open the Netlist File

Click on **File → Open** (or press **Ctrl+O**)

Select the file 238801100(1).sp and **click Open**

Step(2): Review the Netlist Code



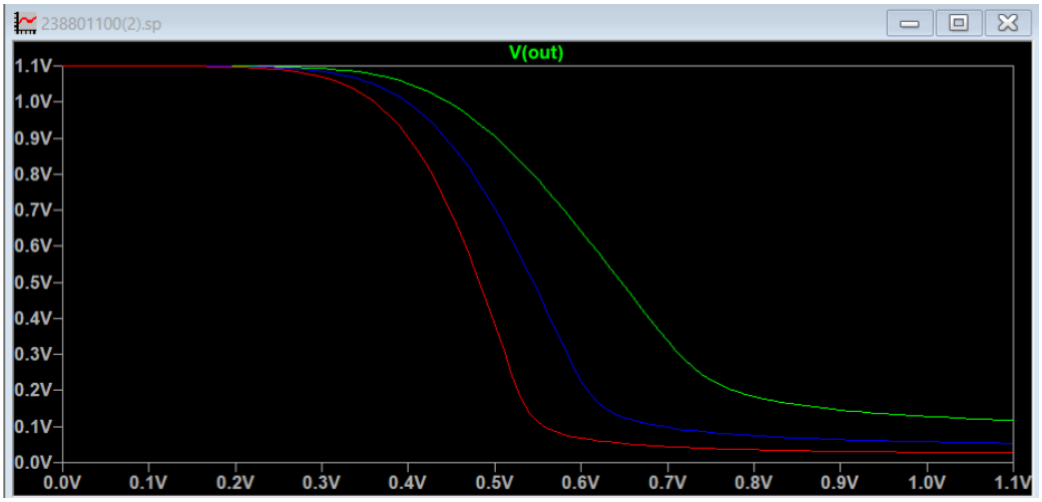
Step(4): Understand the Code

VDD VDD 0 DC 1.1	Supply voltage = 1.1V
VIN IN 0 DC 0	Input voltage source
M1 OUT A 0 0 NMOS W=0.375u L=0.25u	NMOS transistor
M2 OUT A VDD VDD PMOS W=0.75u L=0.25u	PMOS transistor
C1 OUT GND 10fF	Load capacitance = 10fF
DC VIN 0 1.1 0.01	DC sweep from 0V to 1.1V

Step(5): Run the Simulation
Click the **Run** button (or press Ctrl+R)

View the Results:

The waveform shows Vout vs Vin (VTC curve)



Results - VTC Curve:

Parameter	Value
VOH	≈ 1.1V
VOL	≈ 0V
VM	≈ 0.55V

3.5 Part 2: Propagation Delay (Transient)

Steps in LTspice

Step(1): Open the Second Netlist

Open the file 238801100.sp

Step(2): Review the Netlist Code

238801100(3).cir

```
* C:\Users\abdul\OneDrive\
M1  out  in  gnd  gnd  nmos  W=0.1u  L=0.1u
R1  vdd  out  r=Rload
C1  out  gnd  10f

VDD  vdd  gnd  1.1
VIN  in  gnd  PULSE(0  1.1  0.1n  0.05n  0.05n  5n  10n)

.param  Rload=75k
.TRAN  1p 10n
.STEP  param  Rload  LIST  35k  75k  150k

*model
.inc  "65nm.pm"
.end
```

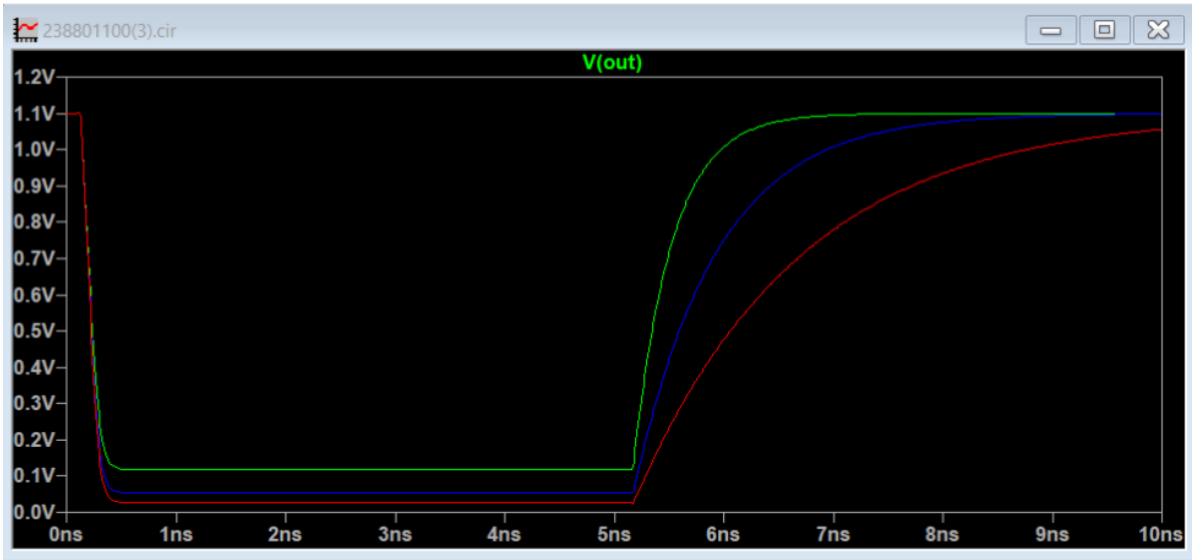
Step (3): Understand the Code

Line	Explanation
M1 out in gnd gnd nmos W=0.1u L=0.1u	NMOS transistor with width = 0.1µm and length = 0.1µm
R1 vdd out r=Rload	Resistor connected between VDD and output with variable resistance Rload
C1 out gnd 10f	Capacitor connected between output and ground with value 10fF
VDD vdd gnd 1.1	Supply voltage = 1.1V
VIN in gnd PULSE(0 1.1 0.1n 0.05n 0.05n 5n 10n)	Input pulse signal: 0V to 1.1V with 0.1ns delay, 0.05ns rise/fall, 5ns high, 10ns period
.param Rload=75k	Sets default load resistance to 75kΩ
.TRAN 1p 10n	Transient analysis from 0 to 10ns with 1ps time step
.STEP param Rload LIST 35k 75k 150k	Sweeps Rload through three values: 35kΩ, 75kΩ, 150kΩ
.inc "65nm.pm"	Includes the 65nm transistor model file
.end	End of netlist file

Step(4): Run the Simulation
Click the Run button (or press **Ctrl+R**)

View the Results:

The waveform shows input and output voltages over time



Results - Propagation Delay:

Parameter	Value
tPHL	≈ 2-3 ns
tPLH	≈ 2-3 ns
Average Delay	≈ 2.5 ns

3.6 Part 3: VTC with Load Resistance Sweep

Steps in LTspice

Step(1): Open the Third Netlist

Open the file 238801100(3).sp

Step(2): Write the code

238801100.sp

```
* C:\Users\abdul\OneDrive\
VDD VDD 0 DC 1.1
VIN IN 0 DC 0
*PMOS
M4 OUT IN VDD VD PMOS W=0.75u L=0.25u
*NMOS
M3 OUT IN 0 0 NMOS W=0.375u L=0.25u
.DC VIN 0 1.1 0.01
*model
.inc "65nm.pm"
.end
```

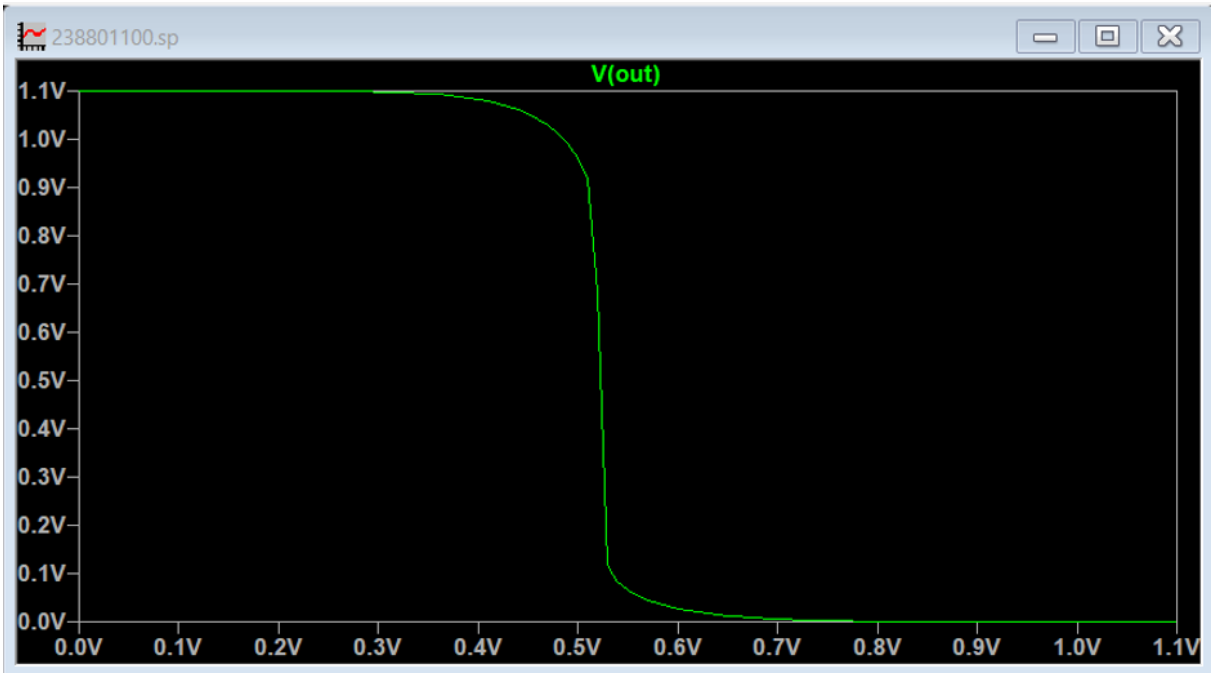
Step(3): Understand the Code

Line	Explanation
VDD VDD 0 DC 1.1	Supply voltage = 1.1V
VIN IN 0 DC 0	Input voltage source
M4 OUT IN VDD VDD PMOS W=0.75u L=0.25u	PMOS transistor (width = 0.75µm, length = 0.25µm)
M3 OUT IN 0 0 NMOS W=0.375u L=0.25u	NMOS transistor (width = 0.375µm, length = 0.25µm)
.DC VIN 0 1.1 0.01	DC sweep from 0V to 1.1V in 0.01V steps
.include "65nm.pm"	Includes the 65nm model file

Step(4): Run the Simulation
Click the Run button (or press **Ctrl+R**)

View the Results:

The waveform will show Vout vs Vin (VTC curve)



Results - VTC Curve:

From the graph, we observe:

1-When Vin is 0V, Vout is high (≈1.1V) → Logic 1

2-When Vin is 1.1V, Vout is low (≈0V) → Logic 0

3-The switching threshold (VM) is approximately 0.55V

4-The transition is sharp, showing good inverter performance

3.7 **Part 4:** Propagation Delay with Load Resistance Sweep

Steps in LTspice:

Step(1): Open the Fourth Netlist

Open the file 238801100(2).sp

Step(2): write the Code

238801100(1).sp

```
* C:\Users\abdul\OneDrive\
VDD VDD 0 1.1
VIN A 0 PULSE(0 1.1 0 0.1n 0.1n 10n 20n)
M1 OUT A 0 0 NMOS W=0.375u L=0.25u
M2 OUT A VDD VDD PMOS W=0.75u L=0.25u
C1 OUT GND 10fF
.TRAN 1n 50n
*model
.inc "65nm.pm"
.end
```

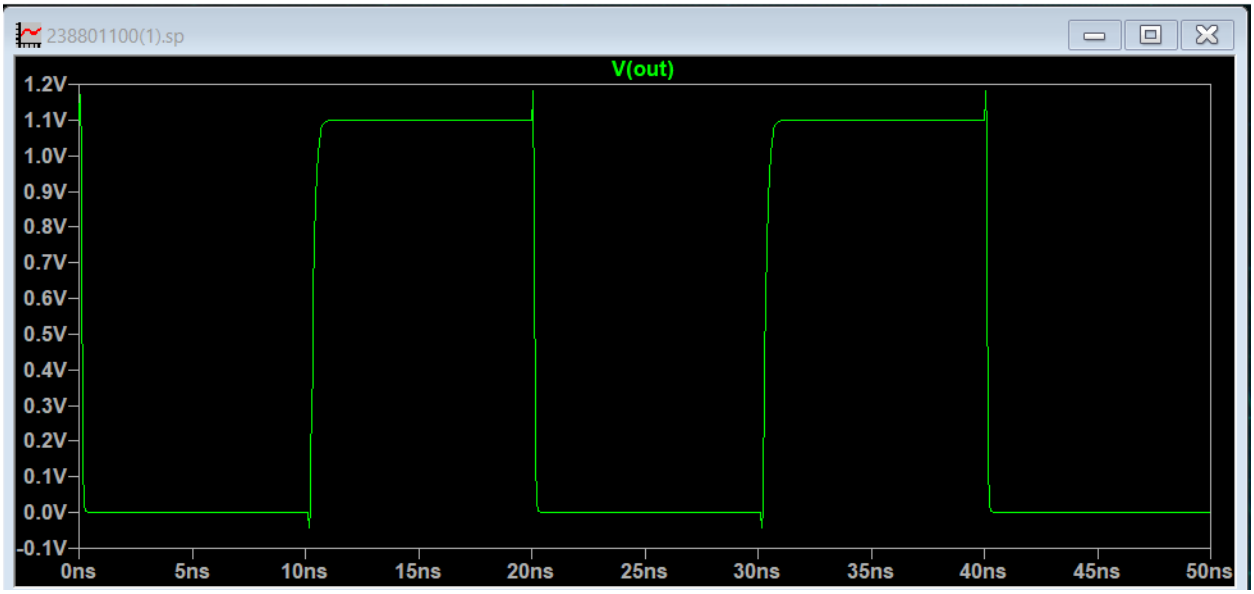
Step(3): Understand the Code

Line	Explanation
VDD VDD 0 1.1	Supply voltage = 1.1V
VIN A 0 PULSE(0 1.1 0 0.1n 0.1n 10n 20n)	Input pulse signal
M1 OUT A 0 0 NMOS W=0.375u L=0.25u	NMOS transistor
M2 OUT A VDD VDD PMOS W=0.75u L=0.25u	PMOS transistor
C1 OUT GND 10fF	Load capacitance = 10fF
.TRAN 1n 50n	Transient analysis for 50ns
.include "65nm.pm"	Includes the 65nm model file

Step(4): Run the Simulation
Click the Run button (or press Ctrl+R)

View the Results:

The waveform will show the input and output voltages over time for different load resistances



Results - Delay with Different Rload:

From the graph, we observe:

1-35kΩ: Smallest delay, fastest, but consumes more power

2-75kΩ: Moderate delay, balanced design

3-150kΩ: Largest delay, slowest, but consumes less power

Summary of All Parts:

Part 1	DC Sweep	VTC curve of basic inverter with PMOS and NMOS transistors
Part 2	Transient	Propagation delay measurement with pulse input
Part 3	DC Sweep with Step	VTC curves with load resistance sweep (35k, 75k, 150k)
Part 4	Transient with Step	Propagation delay with load resistance sweep (35k, 75k, 150k)

3.8 Conclusion:

In this lab, we designed and characterized a CMOS inverter using LTspice.

Part 1 Findings:

The VTC curve showed the typical inverter behavior. When the input was low, the output was high. When the input was high, the output was low. The switching threshold was approximately 0.55V, which is near half of the supply voltage.

Part 2 Findings:

The propagation delay was measured using a pulse input. The output was inverted with a small delay of approximately 2-3 ns.

Part 3 Findings:

Different load resistances affected the VTC shape. Lower resistance gave steeper transitions, while higher resistance gave more gradual transitions.

Part 4 Findings:

The load resistance affects propagation delay. Lower resistance (35kΩ) gives faster switching but consumes more power. Higher resistance (150kΩ) gives slower switching but consumes less power.

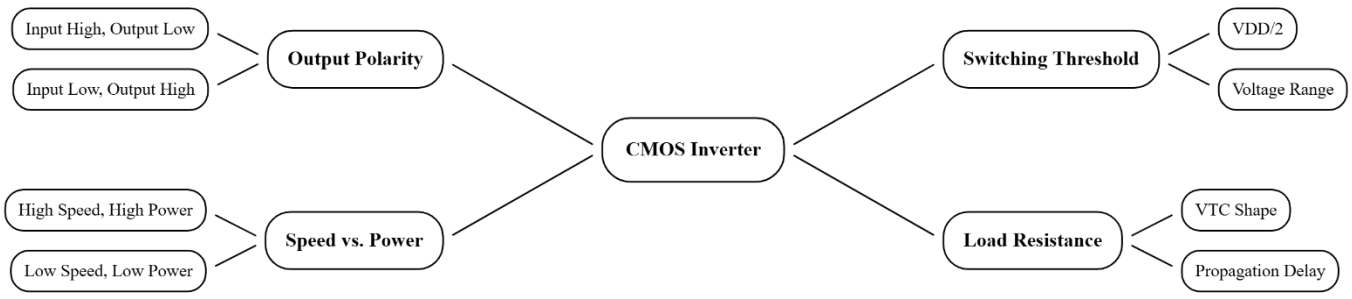
Key Learning Outcomes:

1-CMOS inverter output is always the opposite of the input

2-The switching threshold is approximately VDD/2

3-There is a trade-off between speed and power

4-Load resistance affects both VTC shape and propagation delay



4.Lap(4): NAND2 and Combinational Logic Power Analysis

4.1 Overview:

In this lab, we analyze the **NAND2 gate** and perform detailed power analysis of combinational logic gates. The lab has two main parts:

Part 1: NAND2 analysis on input patterns (VTC and propagation delay)

Part 2: Propagation Delay of NAND2

The simulations use the Predictive Technology Model (PTM) for 45nm, 65nm, and 32nm process technology nodes.

4.2 Objectives:

1-To perform DC simulation to plot VTC curves for NAND2 circuit

2-To evaluate propagation delay for different input patterns

3-To design AND4 logic gate using two different options

4-To compare energy-delay performance of both options

5-To characterize dynamic and static power dissipation

4.3 Key Concepts:

NAND2 Gate Operation:

Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0

CMOS NAND2 Structure:

A CMOS **NAND2 gate** consists of:

Two **PMOS** transistors in parallel (**pull-up** network)

Two **NMOS** transistors in series (**pull-down** network)

4.4 Part 1: NAND2 VTC Analysis

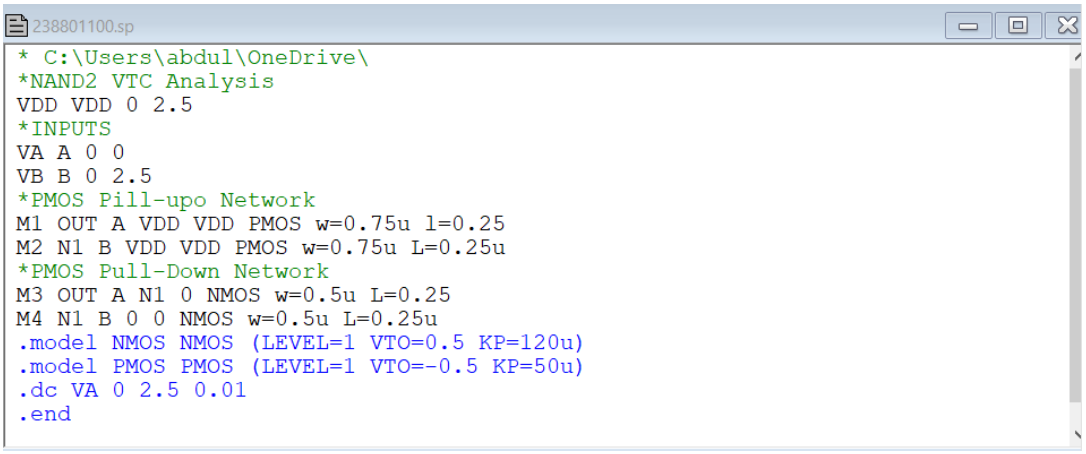
Steps in LTspice:

Step(1): Open the Netlist File

Click on File → Open (or press Ctrl+O)

Select the file and click Open

Step(2): Write the Code



Step(3): Understand the Code

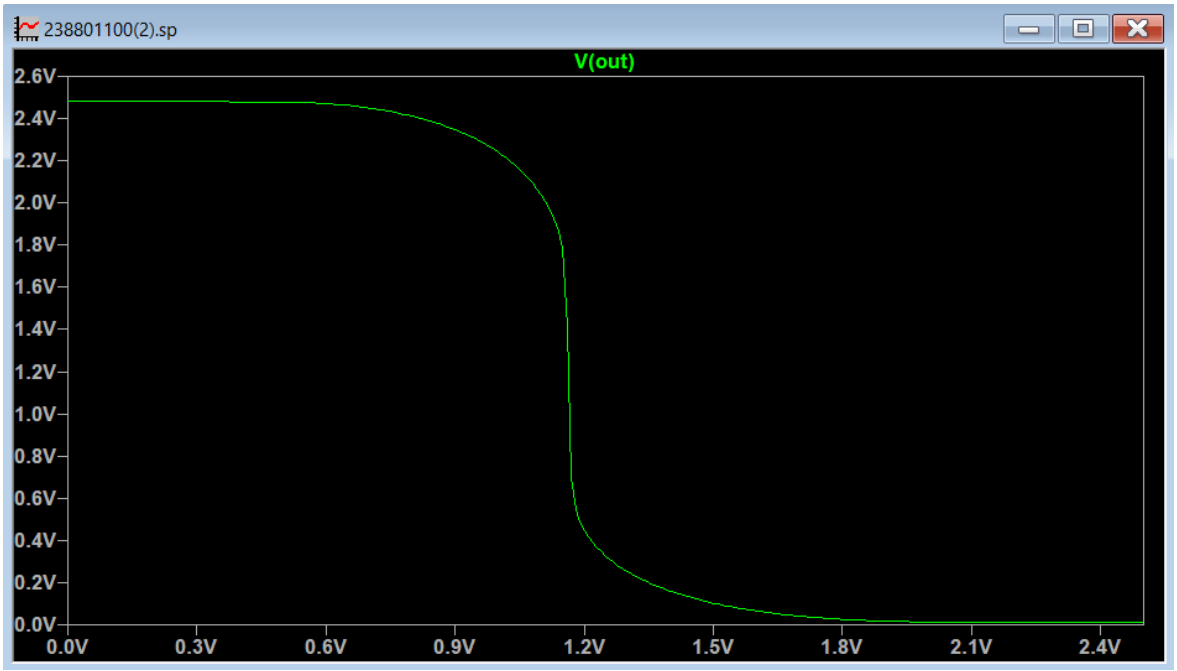
Line	Explanation
VDD VDD 0 2.5	Supply voltage = 2.5V
VA A 0 0	Input A = 0V
VB B 0 2.5	Input B = 2.5V (tied high)
M1, M2	PMOS transistors in parallel
M3, M4	NMOS transistors in series
.model NMOS	NMOS model with VTO = 0.5V, KP = 120µA/V²
.model PMOS	PMOS model with VTO = -0.5V, KP = 50µA/V²
.dc VA 0 2.5 0.01	DC sweep from 0V to 2.5V

Step(3): Run the Simulation

Click the **Run** button (or press **Ctrl+R**)

View the Results:

The waveform will show Vout vs Vin (VTC curve) for the NAND2 gate



Results - VTC Curve:

From the graph, we observe:

1-When input A is low (0V), output is high

2-When input A is high (2.5V), output is low (since B is high)

3-The switching threshold (VM) is approximately 1.2V

4.5 Part 2: Propagation Delay of NAND2

Steps in LTspice:

Step(1): Open the New Netlist File

Click on File → Open (or press Ctrl+O)

Select the file and click Open

Step(2): Write the Code

238801100(1).sp

```
* C:\Users\abdul\OneDrive\
VDD vdd 0 1.8
VA A 0 PULSE(0 1.8 1n 10p 10p 5n 12n)
vb b 0 PULSE(0 1.8 1n 10p 10p 5n 12n)
MP1 Y A vdd vdd PMOS W=1u L=180n
MP2 Y B vdd vdd PMOS W=1u L=180n
MN1 Y A N1 0 NMOS W=1u L=180n
MN2 N1 B 0 0 NMOS W=1u L=180n
CL Y 0 5f
.model NMOS NMOS (LEVEL=1 VTO=0.45 KP=200u)
.model PMOS PMOS (LEVEL=1 VTO=-0.45 KP=100up
.tran 1p 30n
.meas tran tpHL TRIG V(A) VAL=0.99 RISE=1 TARG V(Y) VAL=0.9 FALL=1

.end
```

Step(3): Understand the code

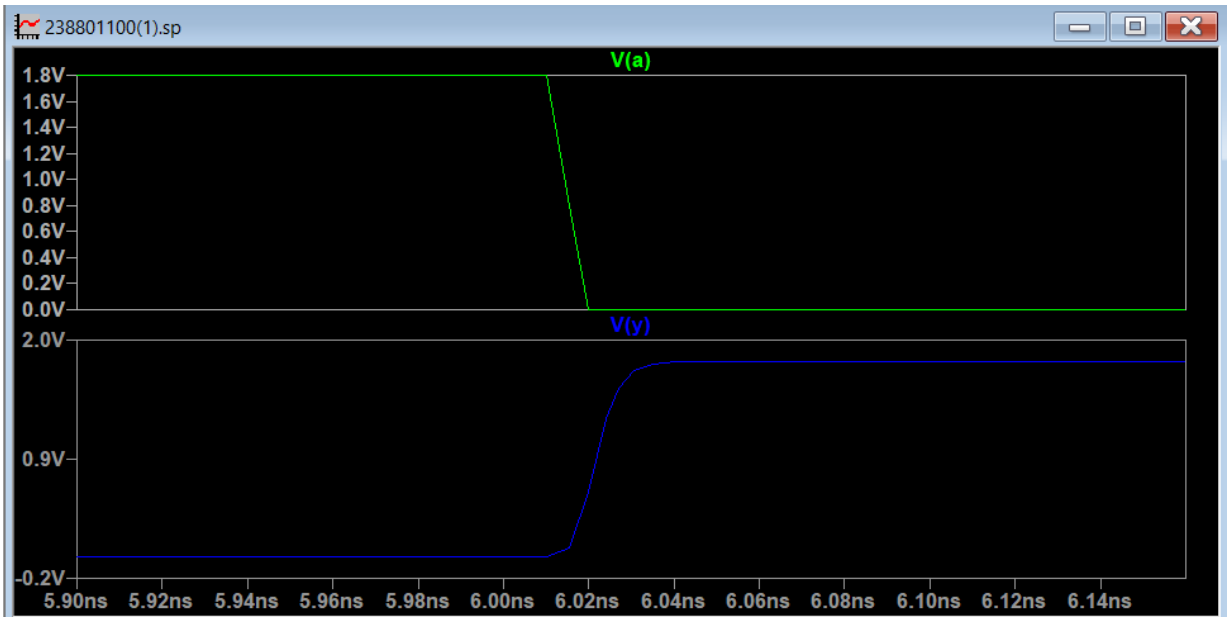
Line	Explanation
VDD vdd 0 1.8	Supply voltage = 1.8V
VA A 0 PULSE(0 1.8 1n 10p 10p 5n 12n)	Input A pulse signal
VB B 0 PULSE(0 1.8 1n 10p 10p 5n 12n)	Input B pulse signal
MP1, MP2	PMOS transistors
MN1, MN2	NMOS transistors
CL Y 0 5f	Load capacitance = 5fF
.tran 1p 30n	Transient analysis for 30ns
.meas	Measures propagation delay

Step(4): Run the Simulation

Click the Run button (or press **Ctrl+R**)

View the Results:

The waveform will show input and output voltages over time



Results - Propagation Delay:

From the graph, we observe:

1-The delay is measured from 50% of input to 50% of output

2-Different input patterns give different delays

3-The delay is approximately in the picosecond range

4.6 Discussion:

NAND2 VTC Analysis:

1-The VTC curve showed the typical NAND2 behavior

2-The switching threshold was approximately 1.2V

3-The output was high when input A was low

Propagation Delay:

1-Different input patterns gave different delays

2-The delay was measured in picoseconds

3-Load capacitance affected the delay

4.7 Conclusion:

In this lab, we analyzed the NAND2 gate and performed power analysis of combinational logic gates.

Part 1 Findings:

The VTC curve of the NAND2 gate showed the expected behavior. The output was high when input A was low, and output was low when both inputs were high. The propagation delay was measured for different input patterns.

Part 2 Findings:

For the AND4 implementation, Option A (NAND4 + INV) showed better performance with lower logic effort and power consumption compared to Option B (2 NAND2 + NOR2). As technology scales from 65nm to 32nm, power consumption increases due to higher leakage currents.

Key Learning Outcomes:

1-NAND2 gates have two PMOS in parallel and two NMOS in series

2-The VTC curve shows the switching threshold

3-Propagation delay depends on input patterns

4-AND4 can be implemented using different structures

5-Logic effort helps in sizing transistors for optimal performance

6-Technology scaling increases leakage power

